

Class 30: Diagonalization

Last class we defined what it meant for two square matrices to be similar

Definition: A is similar to B means there is an invertible matrix P such that $P^{-1} A P = B$ or equivalently, $A = P B P^{-1}$.

If A is similar to a diagonal matrix we say that A is diagonalizable. In this case we can factor A as $P D P^{-1}$. This type of factorization is very nice if you need to raise A to powers.

Big question: Under what conditions will the $n \times n$ matrix A be diagonalizable

Practice: Is the matrix $A = \begin{bmatrix} 5 & -3 & 0 & 9 \\ 0 & 3 & 1 & -2 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 2 \end{bmatrix}$ diagonalizable?

What is the characteristic equation?

What are the eigenvalues?

Find a basis for each eigenspace.

Big theorem: If $\mathbf{v}_1, \dots, \mathbf{v}_r$ are eigenvectors that correspond to distinct eigenvalues for matrix A , then the set $\{\mathbf{v}_1, \dots, \mathbf{v}_r\}$ is linearly independent.

Proof by contradiction:

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Practice: The matrix $A = \begin{bmatrix} 2 & -2 & -2 \\ 3 & -3 & -2 \\ 2 & -2 & -2 \end{bmatrix}$ has eigenvalues -2, -1, and 0.

Find an invertible matrix P and a diagonal matrix D such that $A = P D P^{-1}$.